

Daily Geometry Drill #7 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Section A Rapid Recognition

1. **Answer:** "a circle of radius 5 centred at the origin"

Method: $\sqrt{x^2 + y^2} = 5$ rearranges to $x^2 + y^2 = 25$, a circle of radius 5.

Investigate further: Forgetting to square: quoting radius 25 or a square.

2. **Answer:** $x = 2$

Method: Squaring $x^2 + y^2 = (x - 4)^2 + y^2$: $x^2 = (x - 4)^2$, so $x = 2$.

Investigate further: Giving the single midpoint $(2, 0)$ instead of the whole vertical line.

3. **Answer:** 4

Method: On the x -axis, between A and B we have $PA = x$ and $PB = 6 - x$, so $x = 2(6 - x)$, giving $x = 4$. The Apollonius circle's two x -axis endpoints are $x = 4$ and $x = 12$.

Investigate further: Choosing $x = 2$ (halving 6) or the external endpoint $x = 12$.

4. **Answer:** $y = 1$ or $y = 7$

Method: $|y - 4| = 3$ gives $y = 1$ and $y = 7$, two parallel lines 3 units from $y = 4$.

Investigate further: Keeping only one line $y = 1$ and missing its mirror $y = 7$.

5. **Answer:** $(2, 1)$

Method: A point equidistant from focus $(0, 1)$ and directrix $y = -1$ satisfies $\sqrt{x^2 + (y - 1)^2} = |y + 1|$, i.e. $x^2 = 4y$; $(2, 1)$ works since $4 = 4$.

Investigate further: Naming a point on the y -axis (which only hits the vertex origin for integer y) or mislaying the focus/directrix distance.

6. **Answer:** $(x - 1)^2 + (y - 2)^2 = 9$

Method: Points at distance exactly 3 from $(1, 2)$ form the circle centred there with radius 3.

Investigate further: Writing distance = 3 unsquared, or using the point as an offset in the wrong sign.

7. **Answer:** "a circle of radius 1 centred at the origin"

Method: Q runs over $x^2 + y^2 = 4$ (radius 2); the midpoint $M = OQ/2$ scales radii by $\frac{1}{2}$, giving

$$x^2 + y^2 = 1.$$

Investigate further: Halving the equation's radius linearly to $\sqrt{2}$ or keeping the original radius 2.

8. **Answer:** 1

Method: $x^2 + y^2 = 2x$ rearranges to $(x - 1)^2 + y^2 = 1$, radius 1.

Investigate further: Answering $\sqrt{2}$ (reading the constant 2 as r^2 before completing the square).

9. **Answer:** false

Method: The point $(\sqrt{2}, \sqrt{2})$ lies on $x^2 + y^2 = 4$ yet has $x \neq \pm 2$ and $y \neq \pm 2$; the implication fails in both coordinates.

Investigate further: Mistaking the circle's bounding box for its points; only the four axis points hit ± 2 .

10. **Answer:** $x = 3$

Method: A and B are at equal height $y = 3$, so the perpendicular bisector is the vertical line $x = \frac{1+5}{2} = 3$.

Investigate further: Writing the horizontal line $y = 3$ through both points.

Section B Manipulation Drills

1. **Answer:** A) 4

Method: On the x -axis, $|x| = 2|x - 6|$ gives the endpoints $x = 4$ and $x = 12$; the circle between them has centre $(8, 0)$ and radius 4.

Investigate further: Quoting the centre distance 8 (option B) or the gap 6 (option D).

2. **Answer:** B) $y = 2$

Method: $x^2 + y^2 = x^2 + (y - 4)^2$ gives $y = 2$, the horizontal bisector.

Investigate further: Answering $x = 2$ (wrong orientation) or $y = 4$.

3. **Answer:** B) 4π

Method: The disk has radius 2, so its area is $\pi r^2 = 4\pi$.

Investigate further: Using $\pi r = 2\pi$ (option A) or $r^2 = 16$ (option D).

4. **Answer:** 2π

Method: Circumference $= 2\pi r = 2\pi \cdot 1 = 2\pi$.

Investigate further: Mixing circumference with area (π) or doubling the radius to 4π .

5. **Answer:** C) $x^2 + y^2 < 2$

Method: If $x^2 + y^2 < 1$ then certainly $x^2 + y^2 < 2$; option A is not necessary (a point with $x^2 + y^2 = 0.75$ is inside but not $< \frac{1}{2}$).

Investigate further: Picking a circle whose radius is smaller than the disk (option A) as if it were implied.

6. **Answer:** A) 3

Method: On the x -axis $|x| = 3|x - 2|$ gives internal $x = \frac{3}{2}$ and external $x = 3$; the right endpoint is 3.

Investigate further: Choosing the left endpoint $\frac{3}{2}$ or the y -absent centre guess.

7. **Answer:** B) $x = -\frac{1}{2}$ and $y = \frac{\sqrt{3}}{2}$

Method: This point satisfies $x^2 + y^2 = \frac{1}{4} + \frac{3}{4} = 1$ exactly, so it pins P on the circle. Option A puts P strictly *inside*; C and D do not force $x^2 + y^2 = 1$.

Investigate further: Choosing a strictly-inside condition (option A) because it is a *necessary* symptom of being on the circle.

8. **Answer:** 3

Method: Distance from $(3, 0)$ to the origin is $\sqrt{9} = 3$.

Investigate further: Answering 6 (the segment AB) or 9 (the squared distance).

9. **Answer:** $(2, 3)$

Method: In $x^2 + y^2 + 2gx + 2fy + c = 0$, the centre is $(-g, -f)$; with $2g = -4$ and $2f = -6$ we get $(2, 3)$.

Investigate further: Halving the wrong sign: $(4, 6)$ or negating incorrectly.

10. **Answer:** 2

Method: Completing the square: $(x - 2)^2 + (y + 3)^2 = (-9) + 4 + 9 = 4$, radius 2.

Investigate further: Quoting the constant 9 as the radius or r^2 before shifting.

Section C Substitution & Structure

1. **Answer:** A) $(6, 0)$

Method: Square $PA = 2PB$: $(x + 2)^2 + y^2 = 4((x - 4)^2 + y^2)$. Expand: $x^2 + 4x + 4 + y^2 = 4x^2 - 32x + 64 + 4y^2$, so $3x^2 - 36x + 60 + 3y^2 = 0$, i.e. $x^2 - 12x + 20 + y^2 = 0$, $(x - 6)^2 + y^2 = 16$. Centre $(6, 0)$, radius 4. Option A. The x -axis endpoints from $|x + 2| = 2|x - 4|$ are $x = 2$ (internal) and $x = 10$, midpoint 6.

Investigate further: Quoting the external endpoint $x = 10$ or the midpoint of AB , $(1, 0)$ (options C/D).

2. **Answer:** D) $x = 0$ or $y = 0$

Method: Distance to L_1 is $|x + y|/\sqrt{2}$, to L_2 is $|x - y|/\sqrt{2}$. Equate: $|x + y| = |x - y|$. Squaring: $(x + y)^2 = (x - y)^2$, so $4xy = 0$, i.e. $x = 0$ or $y = 0$, the two coordinate axes (the internal and external angle bisectors of $y = \pm x$). Options A/B give only one bisector; C is not a locus of points equidistant

from lines.

Investigate further: Answering a single line $y = x$ (option A) which is the original L_2 itself, not its bisector; forgetting the modulus gives the second branch.

3. Answer: A)

Method: Fil invokes the internal angle-bisector theorem which requires D to be *interior* to AB . The diagram may place D on the extension of AB beyond B (or beyond A) where the external bisector ratio is $AD/DB = -CA/CB$; without proving convexity/betweenness the betweenness hypothesis is not established. This is the classic “all triangles are isosceles” fallacy class. Hence the argument assumes what it must prove. Option A names the betweenness gap; B miscasts a squared modulus, C misstates the theorem, D claims no flaw.

Investigate further: Misidentifying the gap as a sign/square issue (option B) or declaring the flawed proof sound (option D).

4. Answer: E) none of them individually (all insufficient alone)

Method: For a parallelogram: I. $PQ = QR$ forces a rhombus; a rhombus with angles $60^\circ, 120^\circ$ (e.g. equilateral-parallelogram) is not a square. II. $PR \perp QS$ also forces a rhombus (equivalently to I) but not a square (same counterexample). III. $\angle PQR = 90^\circ$ forces a rectangle; a 2×3 rectangle is not a square. So each alone is insufficient; $I + III$ or $II + III$ together would be sufficient. Hence no single condition suffices, option E. This mirrors TMUA 2020 P2 Q7 answer “none individually sufficient” (official answer H in the 8-option paper, here condensed to E).

Investigate further: Choosing I or III alone (options A/C) because “rhombus” or “right angle” sounds like “square” without checking the other degree of freedom.

5. Answer: E) $p = 2$ or $p \geq 4$

Method: By the sine rule $\sin R/PR = \sin 30^\circ/p$, so $\sin R = 2/p$. The ambiguous case: for $p < 4$ there are potentially two values of R giving two triangles, except when $\sin R = 1$ ($p = 2$) where the two coalesce to a right-angled triangle uniquely determined. For $p \geq PR = 4$ the side opposite the given angle is at least the other given side, so the angle R is uniquely acute; $p = 2$ also unique. Hence exactly $p = 2$ or $p \geq 4$ gives a unique PQ ; all other $p < 4$, $p \neq 2$ give two possible PQ (or none for $p < 2 \sin 30^\circ$). This scales the official TMUA 2018 P2 Q14 answer $p = 1$ or $p \geq 2$ (with $PR = 2$) by factor 2.

Investigate further: Answering $p \geq 4$ alone (option D) and missing the isolated right-angle case $p = 2$; or quoting the unscaled $p = 1$ threshold (option A).

6. Answer: E) 4

Method: Centres $O_1(-3, 1)$, $O_2(8, 1)$, so $d = O_1O_2 = 11$. Radii $r_1 = 5$, $r_2 = 3$. Since $d = 11 > r_1 + r_2 = 8$ the circles are externally separate: 4 common tangents (2 direct, 2 transverse). Compare d with $r_1 + r_2$ and $|r_1 - r_2| = 2$: $d > r_1 + r_2 \Rightarrow 4$, $d = r_1 + r_2 \Rightarrow 3$, $|r_1 - r_2| < d < r_1 + r_2 \Rightarrow 2$, $d = |r_1 - r_2| \Rightarrow 1$, $d < |r_1 - r_2| \Rightarrow 0$. Here $11 > 8$ so 4.

Investigate further: Counting 3 (option D) by mistaking $d = r_1 + r_2$ tangency, or 2 by thinking intersecting.

7. Answer: A) 9

Method: For $PA = 3PB$, $x^2 + y^2 = 9((x-6)^2 + y^2)$ gives $8x^2 - 108x + 8y^2 + 324 = 0$, i.e. $(x - \frac{27}{4})^2 + y^2 =$

$(\frac{9}{4})^2$. Centre $C(\frac{27}{4}, 0) = (6.75, 0)$, radius $r = \frac{9}{4} = 2.25$. The furthest point from the origin on the circle is on the ray OC beyond C : $OC + r = 6.75 + 2.25 = 9$; the nearest is $OC - r = 4.5$. So $\max OP = 9$, option A.

Investigate further: Quoting $OC = 6.75$ (not a listed option) or the nearest distance 4.5/diameter 4.5.

8. Answer: A)

Method: Expand correctly: $x^2 + y^2 = 9((x-2)^2 + y^2)$ gives $x^2 + y^2 = 9x^2 - 36x + 36 + 9y^2$, so $8x^2 - 36x + 8y^2 + 36 = 0$, i.e. $(x - \frac{9}{4})^2 + y^2 = \frac{9}{16}$, centre $(\frac{9}{4}, 0)$, radius $\frac{3}{4}$. Fil's $(x-2)^2 + y^2 = 9$ is the circle centred at $B = (2, 0)$ with radius 3; it uses only PB and ignores A entirely. Option A's locus is the true Apollonius circle; B claims the locus collapses, C claims it is a line ($k = 1$ only), D claims no error.

Investigate further: Accepting the B -centred circle (option D) because its centre “looks like B ” and its radius “is the ratio times 1”.

Section D Challenge Ramp

1. Answer: A) $2\sqrt{2}$

Method: $PA^2 = (x+3)^2 + y^2$, $PB^2 = (x-3)^2 + y^2$. Sum $= 2x^2 + 18 + 2y^2 = 34$, so $x^2 + y^2 = 8$, i.e. $(x-0)^2 + y^2 = 8$, a circle centred at the midpoint $(0, 0)$ with $r^2 = 8$, $r = 2\sqrt{2}$. This is the $PA^2 + PB^2$ locus (Apollonius's theorem generalisation). Check: $PA^2 + PB^2 = 2m^2 + AB^2/2$ with $m = OP$. Option A; B is r^2 , D is total sum.

Investigate further: Answering $\sqrt{8}$ as 8 (option B/D confusion) or forgetting the factor 2 before completing the square.

2. Answer: C) 18

Method: $|x| + |y| = 3$ is four lines $x + y = 3$, $x - y = 3$, etc., forming a diamond with vertices $(3, 0)$, $(0, 3)$, $(-3, 0)$, $(0, -3)$. Its diagonals are 6 and 6, so area $= \frac{1}{2} \cdot 6 \cdot 6 = 18$ (equivalently four right triangles each $\frac{1}{2} \cdot 3 \cdot 3 = 4.5$). For $|x| + |y| = 1$ the area is 2 (TMUA 2021 P1 Q9 answer C), scaling by factor 3^2 gives $2 \cdot 9 = 18$. Option C.

Investigate further: Doubling instead of squaring the scale factor (giving 12/6, options A/B).

3. Answer: C) $x^2 = yz$

Method: Put O at origin, let $SP^2 = x^2 + z^2$, $PQ^2 = x^2 + y^2$, $SQ^2 = (y+z)^2$. Then $\triangle SPQ$ right at P iff $SP^2 + PQ^2 = SQ^2$ (Pythagoras): $(x^2 + z^2) + (x^2 + y^2) = (y+z)^2$, so $2x^2 + y^2 + z^2 = y^2 + 2yz + z^2$, i.e. $2x^2 = 2yz$, $x^2 = yz$. Conversely $x^2 = yz$ reverses the steps; hence the condition is both necessary and sufficient. This is exactly TMUA 2022 P2 Q11 (official answer C). Option C.

Investigate further: Choosing $y = z$ (option D) which only makes the kite symmetric but still an isosceles right possibility, not the product condition.

4. Answer: B) 6

Method: Radical axis: subtract $x^2 + y^2 = 25$ and $(x-4)^2 + y^2 = 9$, i.e. $x^2 - (x-4)^2 = 16$, so $8x - 16 = 16$, $x = 4$. Distance from $O_1(0, 0)$ to the line $x = 4$ is 4. Half-chord $= \sqrt{r_1^2 - d_{\text{axis}}^2} = \sqrt{25 - 16} = 3$, so full chord $PQ = 6$. Check with O_2 : $4 - 4 = 0$, distance 0, half-chord also $\sqrt{9 - 0} = 3$. Option B. The diagram's dashed line is $x = 4$.

Investigate further: Quoting the half-chord 3 (option A) or using $d = O_1O_2 = 4$ directly as the chord.

5. Answer: 12

Method: Apollonius $PA = 2PB$ with $A(0,0), B(6,0)$ is $(x-8)^2 + y^2 = 16$, centre $C(8,0)$, radius $r = 4$. For fixed base $AB = 6$, $[PAB] = \frac{1}{2} \cdot AB \cdot |y_P|$, so maximise $|y_P|$ among points on the circle: $\max |y_P| = r = 4$ (topmost point $(8,4)$). Hence $\max[PAB] = \frac{1}{2} \cdot 6 \cdot 4 = 12$. Check: $(8,4)$ indeed satisfies $(0,4)$? $PA = \sqrt{80} = 4\sqrt{5}$, $PB = \sqrt{20} = 2\sqrt{5}$, ratio 2 correct. Uniqueness up to reflection in x -axis.

Investigate further: Using $r = 16$ (option squared) or $AB = 12$ (diameter) for 24.

Top 5 Patterns

- **Apollonius ratio $PA/PB = k \neq 1$ squares to a circle; read its centre by squaring and completing the square.** The x -axis endpoints from $|x-a| = k|x-b|$ give the diameter; the midpoint is the centre. (see C1, C7, C8, D5)
- **$PA^2 + PB^2$ and $|x| + |y|$ loci are also locus families: sum-of-squares is a centred circle; $|x| + |y| = a$ is a diamond whose area scales as a^2 .** Use AB midpoint and the diagonal product. (see D1, D2)
- **Distinguish *necessary* from *sufficient* and spot the hidden betweenness hypothesis.** A condition is necessary if every square has it; sufficient if it alone forces a square; a proof that needs “ D between A and B ” must prove convexity first. (see C3, C4, D3)
- **The ambiguous case (sine rule) decides uniqueness via $\sin R = PR \sin P/p$; the isolated $\sin R = 1$ case and the $p \geq PR$ regime are the only uniquely-determined ones.** (see C5)
- **Subtract circles for the radical axis; its distance from the centre gives the common chord via $\sqrt{r^2 - d_{\text{axis}}^2}$.** External $d > r_1 + r_2$ counts four tangents. (see C6, D4)

Common Traps

- **Writing an Apollonius locus centred at B (or at the midpoint of AB) instead of completing the squared ratio.** The true centre divides AB externally/internally in ratio $k^2:1$, not $1:1$. (see C1, C7, C8, D5)
- **Collapsing a modulus locus to one branch.** $|x+y| = |x-y|$ has *two* solutions $x=0$ and $y=0$; $|y-4|=3$ has twins $y=1,7$; $|PA|=k|PB|$ has internal *and* external x -axis endpoints. (see A4, C2, C7)
- **Claiming a rhombus or right-angle condition alone forces a square.** Each degree of freedom (side equality vs right angle) must be secured separately. (see C4, D3)
- **Forgetting the scale factor for diamond/bounded loci.** $\text{Area}(|x| + |y| = a) = 2a^2$; doubling a quadruples area, not doubles. (see D2)
- **Quoting a half-chord or a squared radius as the full length.** The distance from the centre to the radical axis is not the chord; $\sqrt{r^2 - d^2}$ is the half-chord, r^2 is not r . (see D1, D4)

“A locus is a sentence the entire family agrees on; a geometric claim is only as good as the condition you

actually invoked.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths